Tobacco Control Data Analysis Blueprint

https://gemini.google.com/app/7ae79071e30a1b17

*User prompt: > **Role & Objective:** > You are an expert data analyst and analytical report designer. I have a one-off task where I need to reproduce the analysis and report structure shown in the attached reference PDF, using data from a newly provided Excel file. > **Inputs Provided:** > 1. **Target Excel Workbook:** Analyze the worksheet at **index 1** (0-based indexing), named **`"HF R37-16 Consolidated Data"`**. > 2. **Reference Analytical PDF:** Serves as the benchmark for the target schema, key metrics/KPIs, charts, tables, narrative structure, and layout. The new dataset shares this schema with minor variations. > > > **Context & Scope:** > This is a **one-time task**, not an ongoing production pipeline. I do not need long-term infrastructure, microservices, or complex workflow automation. I need the fastest, lowest-effort practical path to complete this single report accurately. > **Your Task:** > Provide a clear, actionable, step-by-step procedure and available free online tools/platforms /AI models, AI tools to execute this analysis and generate the final report with minimal manual effort. > **Please include:** > 1. **Schema & Metric Mapping:** How to quickly inspect the sheet, verify schema consistency against the PDF, and identify the required transformations/aggregations. > 2. **Execution Steps (Lowest-Effort Tooling):** Recommended zero-setup tools (e.g., Python scripts using pandas/Jupyter, AI code-interpreter environments, or interactive BI tools) and the exact analytical steps to compute the metrics. > 3. **Report Generation Method:** The most practical way to assemble the final output (e.g., generating Markdown/HTML/PDF via Python, or using AI to draft sections from computed summaries) so it matches the PDF's structure and visual presentation. > 4. **Validation Checklist:** Key sanity checks to ensure the newly calculated numbers align with the methodology demonstrated in the reference PDF.*

Response: Executive Blueprint: Reproducing the Analytical Report

The reference PDF (*Heartfile / ATNiC Tobacco Control Laws Monitoring Report 2025*) follows a structured three-tier assessment framework across **Public Places (Section A)**, **Public Service Vehicles / PSVs (Section B)**, and **Retail Outlets / Points of Sale (Section C)**. The newly provided workbook sheet **`"HF R37-16 Consolidated Data"`** contains raw KoBoToolbox survey data (778 records) covering 7 districts (Bahawalpur, D.G. Khan, Jhang, Mansehra, Mirpurkhas, Sahiwal, and Vehari).

1. Schema & Metric Mapping

A. Data Cleaning & Segmentation

Before running aggregations, two critical normalizations are required:

1. **District Standardization:** Harmonize string casing and typos in column `District` (e.g., merge `'vihari'` into `'Vehari'`).
2. **Category Segmentation:** The `Category of Place` column splits records into three distinct analytical universes:
   - **Public Places ($N \approx 444$):** Filter by `Category of Place == 'Public place'`.
   - **Public Service Vehicles ($N \approx 106$):** Filter by `Category of Place.isin(['Public Service Vehicle (PSV)', 'public_service_vehicle__psv'])`.
   - **Retail Outlets ($N \approx 228$):** Filter by `Category of Place.isin(['Retail outlet /Point of Sale (POS)', 'retail_outlet__point_of_sale__pos'])`.

B. Indicator & Column Crosswalk

Each report table presents two complementary figures: **Observed Prevalence (%)** and **Compliance Rate (%)** (where $\text{Compliance} = 100\% - \text{Violation}$).

1. Public Places (Section A)

Report Indicator	Raw Excel Column Header	Violation Definition	Compliance Definition
'No Smoking' boards displayed	`A3: “No Smoking Area”... boards are visible?`	Value is `'No'`	Starts with `'Yes'`
People found smoking indoors	`A4: PEOPLE found smoking in the indoor area...`	`'Yes'`	`'No'`
Staff found smoking indoors	`A5: STAFF found smoking in the indoor area...`	`'Yes'`	`'No'`
Staff aware of smoking ban	`A6: STAFF is aware of ban on smoking...`	`'No'`	`'Yes'`
Officer aware: duty to stop smoking	`A7: Authorized officer... aware... to stop...`	`'No'`	`'Yes'`
Officer aware: duty to display board	`A8: Authorized officer... aware... to display...`	`'No'`	`'Yes'`

Report Indicator	Raw Excel Column Header	Violation Definition	Compliance Definition
Cigarette butts visible indoors	`A11: Cigarette butts are visible...`	`'Yes'`	`'No'`
Ashtrays / matches visible indoors	`A12: Ashtrays or matches are visible...`	`'Yes'`	`'No'`
Designated smoking area present	`A13` or `A14: Designated smoking area...`	`'Yes'`	`'No'`
Tobacco sold on premises	`A15: Tobacco or nicotine products sold...`	`'Yes'`	`'No'`
Smoke-Free Designation (Composite)	*Calculated boolean rule*	Any violation of criteria	`A3 == 'Yes'` **AND** `A4 == 'No'` **AND** `A5 == 'No'` **AND** `A11 == 'No'` **AND** `A12 == 'No'`

2. Public Service Vehicles / PSVs (Section B)

Report Indicator	Raw Excel Column Header	Violation Definition	Compliance Definition
'No Smoking' sign displayed	`B4: “No Smoking” sign displayed inside PSV?`	Value is `'No'`	Starts with `'Yes'`
Driver smoking at time of visit	`B5: Driver found smoking in public vehicle?`	`'Yes'`	`'No'`
Conductor smoking at time of visit	`B6: Conductor found smoking inside PSV?`	`'Yes'`	`'No'`
Passenger smoking at time of visit	`B7: Passenger found smoking inside PSV?`	`'Yes'`	`'No'`
Cigarette butts visible in PSV	`B10: Cigarette butts are visible inside...`	`'Yes'`	`'No'`
Ashtrays visible in PSV	`B11: Ashtrays are visible inside the PSV?`	`'Yes'`	`'No'`
Staff aware of smoking ban	`B12: PSV staff is aware of ban...`	`'No'`	`'Yes'`
Staff aware: duty to display board	`B13: PSV staff is aware... to display...`	`'No'`	`'Yes'`
Staff aware: duty to stop smoking	`B14: PSV staff is aware... to stop...`	`'No'`	`'Yes'`
Smoke-Free PSV (Composite)	*Calculated boolean rule*	Any violation of criteria	`B4 == 'Yes'` **AND** `B5 == 'No'` **AND** `B6 == 'No'` **AND** `B7 == 'No'`

3. Retail Outlets / Points of Sale (Section C)

Report Indicator	Raw Excel Column Header	Violation Definition	Compliance Definition
Sales to minors taking place	`C4: Sales of tobacco products to minors...`	`'Yes'`	`'No'`
Vendors always verifying age/ID	`C5: Was the vendor checking ID cards...`	`'Never'`, `'Sometimes'`	`'Always'` (Exclude/note N/A if no young buyers visited)
'No Sales to Minors' sign displayed	`C6: Is there a ‘No Sales to Minors...’ sign...`	Value is `'No'`	Starts with `'Yes'`
Cigarettes sold without GHW	`C7: Cigarette packs without graphic health...`	`'Yes'`	`'No'`
Cigarettes sold without excise stamps	`C8: Cigarette packs without excise stamps...`	`'Yes'`	`'No'`

Report Indicator	Raw Excel Column Header	Violation Definition	Compliance Definition
Seller licensed under TVA	`C9: Does the seller have a license under TVA?`	`'No'` / 'Not enforced'	`'Yes'`
Loose cigarettes available for sale	`C10: Loose cigarettes are available for sale?`	`'Yes'`	`'No'`
Placed next to candies/snacks	`C11: Cigarettes are placed next to candies...`	`'Yes'`	`'No'`
Cigarettes advertised indoors	`C12-Cig-indoor-advert...`	`'Yes'`	`'No'`
Cigarettes advertised outdoors	`C12-Cig-outdoor-advert...`	`'Yes'`	`'No'`
Nicotine pouches advertised indoors	`C12-ONP-indoor-advert...`	`'Yes'`	`'No'`
Nicotine pouches advertised outdoors	`C12-ONP-outdoor-advert...`	`'Yes'`	`'No'`
Promotions / discounts offered	`C13... special promotions or discounts...`	`'Yes'`	`'No'`
Giveaways / gifts offered	`C14... giveaways/gifts associated...`	`'Yes'`	`'No'`
Located within 50m of school	`C15: Is the store located within 50m...`	`'Yes'`	`'No'`

> **Column Shift Anomaly Warning:** Inspecting the raw data reveals that some survey responses in columns `C4` and `C5` contain image URLs or filenames (e.g., `*.jpg` or `kf.kobotoolbox.org/...`) due to schema concatenation across survey rounds. When computing metrics, treat any row with non-standard text as missing/NaN or inspect adjacent columns to map the true response.

2. Execution Steps (Lowest-Effort Tooling)

The fastest zero-setup pipeline involves running a lightweight Python script in **Google Colab**, a local **Jupyter Notebook**, or pasting the data into an AI Code Interpreter.

Python Aggregation Script

```
import pandas as pd
import numpy as np

# 1. Load Worksheet
excel_path = "consolidated data.xlsx"
df = pd.read_excel(excel_path, sheet_name="HF R37-16 Consolidated Data")

# 2. Harmonize District Names
df['District'] = df['District'].astype(str).str.strip().str.title()
df['District'] = df['District'].replace({'Vihari': 'Vehari'})

# 3. Define Boolean Normalizer
def is_yes(val):
    if pd.isna(val): return False
    s = str(val).strip().lower()
    return s.startswith('yes') or s == 'true' or s == '1'

def is_no(val):
    if pd.isna(val): return False
    s = str(val).strip().lower()
    return s.startswith('no') or s == 'false' or s == '0'

# 4. Generate Public Place Summary (Example for Section A)
df_pub = df[df['Category of Place'].astype(str).str.contains('Public place', case=False, na=False)].copy()

# Add Smoke-Free Definition Flag
df_pub['is_smoke_free'] = (
    df_pub['A3: “No Smoking Area”, “Smoking is illegal here” or similar boards are visible?'].apply(is_yes) &
    df_pub['A4: PEOPLE found smoking in the indoor area at the time of visit?'].apply(is_no) &
    df_pub['A5: STAFF found smoking in the indoor area at the time of visit?'].apply(is_no) &
    df_pub['A11: Cigarette butts are visible in the indoor area?'].apply(is_no) &
```

```
    df_pub['A12: Ashtrays or matches are visible in the indoor area?'].apply(is_no)
)

# Aggregate Public Places by District
summary_a = df_pub.groupby('District').agg(
    total_sites=('Form Number', 'count'),
    boards_displayed=('A3: “No Smoking Area”, “Smoking is illegal here” or similar boards are visible?', lambda x: f"{x.
    indoor_smoking_observed=('A4: PEOPLE found smoking in the indoor area at the time of visit?', lambda x: f"{x.apply(i
    smoke_free_pct=('is_smoke_free', lambda x: f"{x.mean():.0%}")
).reset_index()

print("Public Places National / District Summary:")
print(summary_a)
```

3. Report Generation Method

To assemble the final PDF without building automated reporting pipelines:

Method: Hybrid AI + HTML-to-PDF Template

1. **Compute Summary Tables:** Run the Python script to output a single Excel workbook containing:
    - **Table 0:** National Overview (National average proportions across all indicators).
    - **Tables 1, 2, 3 per District:** The exact 3-column format shown in the PDF: `[Area of TCL compliance | Percentage of sites (n=X) | Percentage in compliance]`.
2. **Draft Narrative Sections via AI:** Pass each district's aggregated summary table to Gemini or Claude with this standardized prompt:

    > *"You are drafting the district profile for [District Name] in the Heartfile Tobacco Control Laws Implementation Monitoring Report. Based on the attached summary table (n=X public places, n=Y PSVs, n=Z retail outlets), generate:* 1. Key findings (5 bullet points highlighting biggest gaps/strengths) 2. Narrative paragraphs for: 'Smoking in public places', 'Smoking in public service vehicles', and 'Sales, packaging and TAPS restrictions in retail outlets' *Match the exact analytical tone, legal phrasing, and structure of the reference report."*

3. **Assemble in Google Docs or Word:**
    - Copy the generated narrative and paste the formatted tables directly from Excel.
    - Insert field evidence photographs (available in the survey folder or URL links from columns `If yes, add photo Evidence _URL` and `C12-*-evidence`).
    - Export as PDF.

4. Validation & Quality Checklist

Before finalizing the report, run these validation rules against the computed outputs:

- [ ] **Sample Count (n) Integrity:** Verify that total forms per district match survey targets (≈ 100 per district, split roughly into 30–50 public places, 10–35 PSVs, and 20–50 retail outlets).
- [ ] **Prevalence vs. Compliance Polarity:**
    - If People Smoking $= 12\%$, then Indoor Smoking Compliance **must** equal 88%.
    - If Board Displayed $= 8\%$, then Signage Compliance **is** 8% (positive indicator).
- [ ] **Strict Smoke-Free Definition Consistency:**
    - For **Public Places**: A site is smoke-free **only** if signage is visible (`A3`), nobody is smoking (`A4` & `A5`), and no butts/ashtrays are present (`A11` & `A12`).
    - For **PSVs**: A vehicle is smoke-free **only** if signage is visible (`B4`) and no one (driver, conductor, passenger) is smoking (`B5`, `B6`, `B7`).
    - *Sanity check:* The `% Smoke-free` can never exceed the `% 'No Smoking' sign displayed`.
- [ ] **TVA Licensing Status:** For districts where the Tobacco Vendors Act (TVA) has not been notified/enforced, mark the compliance column as `"TVA not enforced in district"` rather than `0%`.
- [ ] **Handling "No Young Buyers":** When calculating vendor age verification in `C5`, report the denominator either as active transactions or footnote: *"No young buyers observed at time of visit"*.

User prompt: although I have an initial Excel file to begin with, I intend to do this with pandas, Python, Jupyter Notebook, and Matplotlib. Is it a good fit? First, tell me in a brief yes, with a brief answer at the beginning of your response, if it's a good fit, and then build up the rest of your argument in your response.

Response: **Yes, this stack is an ideal fit.** It provides the exact balance of programmatic speed, data wrangling control, and reproducible visualization needed to replicate this specific report without the overhead of enterprise BI tools or custom software infrastructure.

Why Python, Pandas, Jupyter, and Matplotlib Fit This Project

1. Precision Handling of Survey "Gotchas" (Pandas)

Raw survey exports from KoBoToolbox are rarely analysis-ready:

- **Inconsistent Category Labels:** The sheet contains split category strings (e.g., `'Public Service Vehicle (PSV)'` vs. `'public_service_vehicle__psv'`, and `'Retail outlet /Point of Sale (POS)'` vs. `'retail_outlet__point_of_sale__pos'`). In Excel, writing nested `COUNTIFS` across multiple variants across 178 columns is brittle and error-prone. In Pandas, this is handled with a single vector mapping:

  ```
  df['Category'] = df['Category of Place'].replace({
      'public_service_vehicle__psv': 'PSV',
      'Public Service Vehicle (PSV)': 'PSV',
      'retail_outlet__point_of_sale__pos': 'Retail',
      'Retail outlet /Point of Sale (POS)': 'Retail',
      'Public place': 'Public Place'
  })
  ```

- **Column Shift & Stray URL Cleanup:** Inspecting columns `C4` and `C5` reveals image URLs and file attachment paths mixed into boolean response columns. A Pandas script easily filters or coercively casts responses (e.g., standardizing on `.str.startswith('Yes')`), preventing skewed denominators.

- **Typo Harmonization:** A one-line `.replace({'vihari': 'Vehari'})` permanently standardizes district aggregation boundaries.

2. Vectorized Multi-Condition Logic for Composite KPIs

In the reference PDF, key benchmark indicators like **"Smoke-Free Status"** require multi-column boolean rules evaluated simultaneously across survey sections:

- **Smoke-Free Public Place Criteria:** Requires signage present (`A3 == Yes`), no people smoking (`A4 == No`), no staff smoking (`A5 == No`), no visible butts (`A11 == No`), and no visible ashtrays (`A12 == No`).

- **Smoke-Free PSV Criteria:** Requires signage present (`B4 == Yes`) and driver, conductor, and passenger all not smoking (`B5 == No`, `B6 == No`, `B7 == No`).

Calculating these row-by-row in Excel across hundreds of rows and multiple tabs invites formula discrepancies. In Pandas, boolean masking creates clean, verifiable logical flags in seconds:

```
df_pub['is_smoke_free'] = (
    df_pub['A3'].str.startswith('Yes', na=False) &
    (df_pub['A4'] == 'No') &
    (df_pub['A5'] == 'No') &
    (df_pub['A11'] == 'No') &
    (df_pub['A12'] == 'No')
)
```

3. Pixel-Accurate Replication of PDF Visualizations (Matplotlib)

The reference analytical PDF relies on two standard chart archetypes:

1. **Horizontal Bar Charts (National Compliance):** Figure 1, Figure 3, and Figure 5 show indicators on the vertical axis, a 0%–120% horizontal grid, and solid blue bars (`#3366cc` / `#2b5b84` palette). Matplotlib's `plt.barh()` replicates these charts with exact tick intervals and labels.
2. **Vertical Column Charts (District Comparisons):** Figure 2 and Figure 4 show district-by-district smoke-free percentages with 45-degree angled x-axis tick labels. Matplotlib handles label rotation and custom axis limits (`ylim(0, 0.6)`) natively.

By using Matplotlib scripts, you generate all five report figures directly to `.png` at 300 DPI, styled consistently with the benchmark document.

4. Iterative Exploration and Rapid Export (Jupyter Notebook)

- **Block-by-Block Execution:** A single notebook allows you to clean data in Cell 1, run national aggregations in Cell 2, inspect district subsets in Cell 3, and generate charts in Cell 4.

- **Direct Table Export:** You can format district compliance summaries directly as Markdown or export all district tables into a multi-sheet Excel file or formatted HTML via `to_excel()` and `to_html()`, ready to paste into your final report document.

Suggested Notebook Workflow Architecture

```
notebook.ipynb
│
├── Cell 1: Environment & Data Ingestion (Read 'HF R37-16 Consolidated Data')
├── Cell 2: Cleaning & Schema Normalization (District names, response casing, category consolidation)
```

```
├── Cell 3: Feature Engineering (Composite flags: is_smoke_free_public, is_smoke_free_psv)
├── Cell 4: National Aggregations (Overall % compliance for Sections A, B, C)
├── Cell 5: Matplotlib Visualizations (Save Figures 1–5 as high-res PNGs)
├── Cell 6: District-by-District Tables Loop (Iterate through 7 districts, output 3 tables per district)
└── Cell 7: Summary Validation Checks (Assert compliance + violation == 100%, check sample sizes)
```

This setup keeps your workspace lightweight, fully reproducible, and fast to execute.

User prompt: all right, do you need to rewrite the data analysis blueprint once again, or is the one you produced earlier good enough? I think producing a new one would be good enough to not only capture whatever you produced in the previous blueprint but also capture the fact that Python, pandas, Jupyter Notebook, and Matplotlib are on the charts for doing this.

Response: This updated, end-to-end data analysis blueprint integrates the specific Python, pandas, Jupyter Notebook, and Matplotlib workflow directly against worksheet index 1 (`"HF R37-16 Consolidated Data"`).

Tobacco Control Laws (TCL) Monitoring Analysis Blueprint

Environment: Jupyter Notebook / Python 3.10+ (`pandas`, `matplotlib`, `numpy`, `openpyxl`)

Data Input: Worksheet index `1` (`"HF R37-16 Consolidated Data"`, 778 records, 178 columns)

Target Output: Exact reproduction of the benchmark analytical report tables, KPIs, national summary charts, and narrative findings.

1. Schema & Metric Mapping Architecture

The survey divides venues into three distinct analytical universes based on `Category of Place`:

- **Public Places (Section A):** $N = 444$
- **Public Service Vehicles / PSVs (Section B):** $N = 106$
- **Retail Outlets / Points of Sale (Section C):** $N = 228$

Exact Column Mapping & Mathematical Logic

The benchmark report presents two percentage columns for each indicator: **Observed %** and **Compliance %**. In legal monitoring, indicators are either *positive duties* (e.g., boards displayed) or *prohibitions* (e.g., indoor smoking, sales to minors).

$$\text{Compliance Rate (Positive Duty)} = \frac{N_{\text{Yes}}}{N_{\text{Total}}} \times 100\%$$

$$\text{Compliance Rate (Prohibition)} = \left(1 - \frac{N_{\text{Violation}}}{N_{\text{Total}}}\right) \times 100\% = \frac{N_{\text{No}}}{N_{\text{Total}}} \times 100\%$$

```
+-----------------------------------------------------------------------------------------------------------------------
|                                                INDICATOR MAPPING DICTIONARY
+----------------------+-------------------------------------------------------------------------+------------+--------------
| Metric Label         | Exact Source Column Header in Worksheet                                 | Direction  | Compliance Ru
+----------------------+-------------------------------------------------------------------------+------------+--------------
| PUBLIC PLACES (SEC A)|                                                                         |            |
| - Signage visible    | A3: “No Smoking Area”, “Smoking is illegal here” or similar boards      | Positive   | Starts with Y
| - People smoking     | A4: PEOPLE found smoking in the indoor area at the time of visit?       | Violation  | Value == 'No'
| - Staff smoking      | A5: STAFF found smoking in the indoor area at the time of visit?        | Violation  | Value == 'No'
| - Staff aware of ban | A6: STAFF is aware of ban on smoking in public place?                   | Positive   | Value == 'Yes
| - Officer aware stop | A7: Authorized officer... aware... to stop persons from smoking         | Positive   | Value == 'Yes
| - Officer aware sign | A8: Authorized officer... aware... to display ‘no smoking’ sign         | Positive   | Value == 'Yes
| - Butts visible      | A11: Cigarette butts are visible in the indoor area?                    | Violation  | Value == 'No'
| - Ashtrays visible   | A12: Ashtrays or matches are visible in the indoor area?                | Violation  | Value == 'No'
| - Smoking area       | A13: Is there a “designated smoking area” in the public place?          | Violation  | Value == 'No'
| - Tobacco sold       | A14: Are there any tobacco or nicotine products being SOLD...           | Violation  | Value == 'No'
|                      |                                                                         |            |
| PSVs (SECTION B)     |                                                                         |            |
| - Signage inside PSV | B4: “No Smoking” sign displayed inside PSV?                             | Positive   | Starts with Y
| - Driver smoking     | B5: Driver found smoking in public vehicle?                             | Violation  | Value == 'No'
| - Conductor smoking  | B6: Conductor found smoking inside PSV?                                 | Violation  | Value == 'No'
| - Passenger smoking  | B7: Passenger found smoking inside PSV?                                 | Violation  | Value == 'No'
| - Butts visible      | B10: Cigarette butts are visible inside the PSV?                        | Violation  | Value == 'No'
| - Ashtrays visible   | B11: Ashtrays are visible inside the PSV?                               | Violation  | Value == 'No'
| - Staff aware ban    | B12: PSV staff is aware of ban on smoking in PSVs?                      | Positive   | Value == 'Yes
| - Staff aware sign   | B13: PSV staff is aware... to display ‘no smoking’ signboard?           | Positive   | Value == 'Yes
```

```
| - Staff aware stop   | B14: PSV staff is aware... to stop persons from smoking in vehicle | Positive   | Value == 'Yes
|                      |                                                                    |            |
| RETAIL (SECTION C)   |                                                                    |            |
| - Sales to minors    | C4: Sales of tobacco products to minors... taking place at visit   | Violation  | Value == 'No'
| - ID card checks     | C5: Was the vendor checking ID cards of YOUNG buyers...            | Positive   | Value=='Alway
| - 'No Sales' sign    | C6: Is there a ‘No Sales to Minors/Persons Under 18’ sign...       | Positive   | Starts with Y
| - No GHW on packs    | C7: Cigarette packs without graphic health warning available...    | Violation  | Value == 'No'
| - No excise stamp    | C8: Cigarette packs without excise stamps available for sale?      | Violation  | Value == 'No'
| - TVA license        | C9: Does the seller have a license under TVA?                      | Positive   | Value == 'Yes
| - Loose cigarettes   | C10: Loose cigarettes are available for sale?                     | Violation  | Value == 'No'
| - Next to candies    | C11: Cigarettes are placed next to candies, snacks...              | Violation  | Value == 'No'
| - Cig indoor ads     | C12-Cig-indoor-advert:  Are Cigarette brands advertized indoors?   | Violation  | Value == 'No'
| - Cig outdoor ads    | C12-Cig-Outdoor-advert: Are Cigarette brands advertized Outdoors?  | Violation  | Value == 'No'
| - ONP indoor ads     | C12-ONP-indoor-advert: Are Nicotine Pouches advertized indoors?    | Violation  | Value == 'No'
| - ONP outdoor ads    | C12-ONP-Outdoor-advert: Are Nicotine Pouches advertized outdoors?  | Violation  | Value == 'No
| - Promo / discounts  | C13: Special promotions or discounts offered...                    | Violation  | Value == 'No'
| - Giveaways / gifts  | C14: Any giveaways/gifts associated with tobacco products          | Violation  | Value == 'No'
| - Within 50m school  | C15: Is the store located within 50m of an educational institution | Violation  | Value == 'No'
+----------------------+--------------------------------------------------------------------+------------+--------------
```

Composite Smoke-Free Criteria Definitions

- **Smoke-Free Public Place:** Requires signage present (`A3 == Yes`) **AND** no indoor active smoking (`A4 == No` and `A5 == No`) **AND** no evidence of smoking (`A11 == No` and `A12 == No`).
- **Smoke-Free PSV:** Requires signage present (`B4 == Yes`) **AND** no active smoking in the vehicle (`B5 == No`, `B6 == No`, and `B7 == No`).

2. Jupyter Notebook Execution Workflow

A clean, modular 5-cell structure executes the entire pipeline end-to-end.

```
Jupyter Notebook Pipeline
│
├── [Cell 1] Imports & Schema Normalization (Regex cleaning, category merging, district casing)
├── [Cell 2] Boolean Feature Engineering (Vectorized positive duties, violations, & composite flags)
├── [Cell 3] District & National Table Generation (Exporting Tables 1–48 in reference format)
├── [Cell 4] Matplotlib Visualizations (Generating Figures 1–5 at 300 DPI matching PDF style)
└── [Cell 5] Validation Suite & Markdown Exporter (Automated sanity checks & drafting summaries)
```

Cell 1: Environment & Schema Normalization

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

# Load sheet at index 1
excel_file = "consolidated data.xlsx"
df_raw = pd.read_excel(excel_file, sheet_name=1)

# Standardize District names
df = df_raw.copy()
df['District'] = df['District'].astype(str).str.strip().str.title()
df['District'] = df['District'].replace({'Vihari': 'Vehari', 'Dg Khan': 'DG Khan'})

# Standardize Category of Place
category_map = {
    'Public place': 'Public Place',
    'Public Service Vehicle (PSV)': 'PSV',
    'public_service_vehicle__psv': 'PSV',
    'Retail outlet /Point of Sale (POS)': 'Retail Outlet',
    'retail_outlet__point_of_sale__pos': 'Retail Outlet'
}
df['Clean_Category'] = df['Category of Place'].map(category_map)
print("Standardized Counts:\n", df.groupby(['District', 'Clean_Category']).size().unstack().fillna(0))
```

Cell 2: Feature Engineering & Robust Parsers

```
def check_yes(series):
    return series.astype(str).str.strip().str.lower().str.startswith('yes')

def check_no(series):
    return series.astype(str).str.strip().str.lower().str.startswith('no')
```

```
# Section A (Public Places)
df_pub = df[df['Clean_Category'] == 'Public Place'].copy()
df_pub['flag_sign'] = check_yes(df_pub['A3: “No Smoking Area”, “Smoking is illegal here” or similar boards are visible?'
df_pub['flag_no_people_smoking'] = check_no(df_pub['A4: PEOPLE found smoking in the indoor area at the time of visit?'])
df_pub['flag_no_staff_smoking'] = check_no(df_pub['A5: STAFF found smoking in the indoor area at the time of visit?'])
df_pub['flag_no_butts'] = check_no(df_pub['A11: Cigarette butts are visible in the indoor area?'])
df_pub['flag_no_ashtrays'] = check_no(df_pub['A12: Ashtrays or matches are visible in the indoor area?'])

# Composite Smoke-Free Public Place
df_pub['is_smoke_free'] = (
    df_pub['flag_sign'] &
    df_pub['flag_no_people_smoking'] &
    df_pub['flag_no_staff_smoking'] &
    df_pub['flag_no_butts'] &
    df_pub['flag_no_ashtrays']
)

# Section B (PSVs)
df_psv = df[df['Clean_Category'] == 'PSV'].copy()
df_psv['flag_sign'] = check_yes(df_psv['B4: “No Smoking” sign displayed inside PSV?'])
df_psv['flag_no_driver_smoking'] = check_no(df_psv['B5: Driver found smoking in public vehicle?'])
df_psv['flag_no_conductor_smoking'] = check_no(df_psv['B6: Conductor found smoking inside PSV?'])
df_psv['flag_no_passenger_smoking'] = check_no(df_psv['B7: Passenger found smoking inside PSV?'])

# Composite Smoke-Free PSV
df_psv['is_smoke_free'] = (
    df_psv['flag_sign'] &
    df_psv['flag_no_driver_smoking'] &
    df_psv['flag_no_conductor_smoking'] &
    df_psv['flag_no_passenger_smoking']
)

# Section C (Retail)
df_ret = df[df['Clean_Category'] == 'Retail Outlet'].copy()
```

Cell 3: District Table Generator Loop

```
def generate_district_tables(district_name):
    dp = df_pub[df_pub['District'] == district_name]
    db = df_psv[df_psv['District'] == district_name]
    dr = df_ret[df_ret['District'] == district_name]

    np_cnt, nb_cnt, nr_cnt = len(dp), len(db), len(dr)

    # 1. Public Places Table
    table_a = pd.DataFrame([
        ("Public places with 'No Smoking' boards displayed", f"{dp['flag_sign'].mean():.0%}", f"{dp['flag_sign'].mean():
        ("Public places with people found smoking in indoor area", f"{(~dp['flag_no_people_smoking']).mean():.0%}", f"{d
        ("Public places with staff found smoking in indoor area", f"{(~dp['flag_no_staff_smoking']).mean():.0%}", f"{dp[
        ("Public places where staff aware of smoking ban", f"{check_yes(dp['A6: STAFF is aware of ban on smoking in publ
        ("Public places where butts are visible", f"{(~dp['flag_no_butts']).mean():.0%}", f"{dp['flag_no_butts'].mean():
        ("Public places where ashtrays/matches visible", f"{(~dp['flag_no_ashtrays']).mean():.0%}", f"{dp['flag_no_ashtr
    ], columns=[f"Area of TCL compliance in public places", f"Percentage of public places (n={np_cnt})", "Percentage in

    return table_a

# Example execution for Bahawalpur
print(generate_district_tables('Bahawalpur'))
```

3. Matplotlib Visualizations (Matching Report Figures)

The PDF uses horizontal bar charts for national compliance metrics (Figures 1, 3, 5) and vertical column charts for cross-district smoke-free comparisons (Figures 2, 4). Matplotlib settings replicate the document's styling (`#2b5b84` bar color, dashed horizontal grids, and rotated labels).

Figure 1 & 3: Horizontal National Compliance Bar Charts

```
def plot_horizontal_compliance(indicators, percentages, title, filename):
    plt.figure(figsize=(10, 6), dpi=300)
    y_pos = np.arange(len(indicators))

    plt.barh(y_pos, percentages, color='#2b5b84', height=0.6)
    plt.yticks(y_pos, indicators, fontsize=9)
    plt.xlim(0, 1.2)
```

```
    plt.gca().xaxis.set_major_formatter(plt.FuncFormatter(lambda y, _: '{:.0%}'.format(y)))
    plt.gca().invert_yaxis()  # Top-down order matching PDF
    plt.grid(axis='x', linestyle='--', alpha=0.5)
    plt.title(title, fontsize=12, pad=15, fontweight='bold')
    plt.tight_layout()
    plt.savefig(filename)
    plt.show()

# Example: Public Places National Compliance
indicators_fig1 = [
    "No Smoking boards displayed",
    "No people found smoking in indoor area",
    "No staff found smoking in indoor area",
    "Staff aware of ban on smoking in public places",
    "Authorized officer aware of duty to stop people smoking",
    "Authorized officer aware of duty to display boards",
    "No cigarette butts visible in indoor area",
    "No ashtrays or matches visible in indoor area",
    "No designated smoking area",
    "No tobacco or nicotine products sold on premises"
]
# Replace with actual .mean() aggregations across df_pub
national_pub_vals = [0.11, 0.96, 0.99, 0.92, 0.89, 0.64, 0.92, 0.97, 0.99, 0.97]
plot_horizontal_compliance(indicators_fig1, national_pub_vals,
                           "Percentage of public places in compliance with TCLs - national", "Figure_1.png")
```

Figure 2 & 4: Vertical Cross-District Smoke-Free Column Charts

```
def plot_district_smoke_free(districts, percentages, title, filename):
    plt.figure(figsize=(11, 5), dpi=300)
    x_pos = np.arange(len(districts))

    plt.bar(x_pos, percentages, color='#2b5b84', width=0.45)
    plt.xticks(x_pos, districts, rotation=45, ha='right', fontsize=9)
    plt.ylim(0, 0.6)
    plt.gca().yaxis.set_major_formatter(plt.FuncFormatter(lambda y, _: '{:.0%}'.format(y)))
    plt.grid(axis='y', linestyle='--', alpha=0.5)
    plt.title(title, fontsize=12, pad=15, fontweight='bold')
    plt.tight_layout()
    plt.savefig(filename)
    plt.show()

# Run directly on df_pub grouped by District
district_series = df_pub.groupby('District')['is_smoke_free'].mean()
plot_district_smoke_free(district_series.index.tolist(), district_series.values,
                         "Smoke-free public places (% of total) by district - national", "Figure_2.png")
```

4. Report Assembly & Narrative Generation

For a single, one-off report, avoid building programmatic Word/PDF rendering pipelines. Combine Jupyter table exports with AI narrative drafting:

1. **Dump Structured Tables to Excel:** Export all computed tables into a formatted Excel sheet via `pd.ExcelWriter("Report_Tables_Consolidated.xlsx")`.
2. **Draft District Text via AI:** Feed each district's completed numbers into an LLM using the benchmark prompt:

   > *"Draft the district analysis for [District Name] using these exact computed figures: n=X public places (Y% smoke-free), n=A PSVs (B% smoke-free), n=P retail outlets (Q% loose sales, R% near schools). Structure into: 1. Key Findings (5 bullet points); 2. Smoking in public places; 3. Smoking in PSVs; 4. Sales, packaging and TAPS restrictions. Match the legal tone and vocabulary of the Heartfile TCL 2025 report."*

3. **Assemble in Google Docs / Word:**
   - Paste the narrative sections directly into a document template matching the reference PDF typography (Calibri/Arial, dark headers, numbered tables).
   - Paste the formatted Excel summary tables.
   - Insert the high-resolution Matplotlib PNG charts (Figures 1 to 5).
   - Retrieve field photos using the survey URLs in columns `If yes, add photo Evidence _URL` and `C12-*-evidence` and place them alongside the corresponding district sections.

5. Strict Analytical Validation Checklist

Run this programmatic checklist in Jupyter before finalizing any text or tables:

```
# Validation 1: Universal Balance Check
# In every row of every table, Violation % + Compliance % must equal 100%
assert ((dp['flag_no_people_smoking'].mean() + (~dp['flag_no_people_smoking']).mean()) - 1.0) < 1e-5

# Validation 2: Upper Bound Invariant for Composite Indicators
# Smoke-free places can NEVER exceed the percentage of venues displaying boards
assert df_pub.groupby('District')['is_smoke_free'].mean().le(
       df_pub.groupby('District')['flag_sign'].mean()
).all(), "Error: Smoke-free % exceeds board display %!"

# Validation 3: PSV Smoke-Free Invariant
assert df_psv.groupby('District')['is_smoke_free'].mean().le(
       df_psv.groupby('District')['flag_sign'].mean()
).all(), "Error: Smoke-free PSV % exceeds PSV board display %!"

# Validation 4: Sample Size Invariant
district_counts = df.groupby(['District', 'Clean_Category']).size()
assert district_counts.min() >= 5, "Warning: A district has fewer than 5 observations in a category!"
```

- [ ] **TVA Enforcement Distinction:** Ensure districts where the Tobacco Vendors Act is not legally notified are explicitly labeled `"TVA not enforced in district"` in Table 3 / retail tables rather than reporting `0%` compliance.
- [ ] **ID Verification Denominator:** When computing `C5` (age verification), verify whether the surveyor selected `"Not Applicable/ No Young buyers visited"`. If no young buyers were present during visits, report the compliance cell as `"No young buyers at time of visit"` rather than an inaccurate `0%`.
